ANSI/IES LM-45-20(R2023)

Illuminating
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APPROVED METHOD:
ELECTRICAL AND PHOTOMETRIC MEASUREMENT OF GENERAL SERVICE INCANDESCENT FILAMENT LAMPS

AN AMERICAN NATIONAL STANDARD

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has been approved by IES.
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should be directed to IES.

Prepared by the
IES Testing Procedures Committee

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Foreword

This Approved Method is a revision of *IES LM-45-15, IES Approved Method: Electrical and Photometric Measurements of General Service Incandescent Filament Lamps*. References were updated.

1.0 Introduction and Scope

1.1 Introduction

Incandescent filament lamps produce radiant power as a result of electric current passing through a tungsten filament, which is surrounded by an inert atmosphere or vacuum within a glass or quartz envelope. Some lamps contain halogen gasses, which are employed to maintain a clean bulb wall. Such lamps may also employ bulb coatings that redirect infrared energy back to the filament to improve efficacy or to filter radiation for color control.

As long as the filament remains intact, current will flow, heating the filament to incandescence. Since the desired incandescence occurs at high filament temperatures (in excess of 2,000 °C), the surface of the tungsten filament is continually vaporized during lamp operation. As a result, the filament wire diameter is non-uniformly decreased along its length until, at some point, the high current density causes excessive local heating and vaporization, which causes the filament to fail. The rate of evaporation is dependent on the local filament temperature, and gas density and pressure.

Incandescent filament lamps are typically affected by variables such as operating cycle, conditions imposed by the fixture, orientation and vibration. In general, the test conditions should not diverge widely from conditions of service. Practical considerations require that any test conditions and programs be designed to give comparable results when used by various laboratories. The recommendations of this IES Approved Method have been made with these objectives in mind.

For special purposes, it may be desirable to determine the characteristics of lamps when they are operated at other than the standard conditions described in this Approved Method. Where this is done, such results are meaningful only for the particular conditions under which they were obtained. All such nonstandard operating conditions shall be stated in the test report.

The photometric information usually required is total luminous flux (lumens), luminous intensity (candelas) in one or more directions, and chromaticity. For the purposes of this Approved Method, the determination of these data will be considered *photometric measurements*.

The electrical characteristics usually measured are lamp current, lamp voltage, and lamp power. Incandescent filament lamps are usually measured on direct current (DC), and the power can be calculated from voltage and current. For the purpose of this Approved Method, the determination of these data will be considered *electrical measurements*.

1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements of the electrical and photometric characteristics of general service incandescent filament lamps under standard conditions. Measurement of incandescent reflector lamps is not included in this Approved Method.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Feb 25).

2.2 ANSI/IES LM-54-20

Illuminating Engineering Society. Approved Method: Guide to Lamp Seasoning. New York: IES; 2020.

2.3 ANSI/IES LM-78-20

Illuminating Engineering Society. Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.

3.0 Nomenclature and Definitions

The units of electrical measurement used in this Approved Method are the volt, the ampere, and the watt. The ambient temperature measurement unit is the degree Celsius (degree Fahrenheit). The units of photometric measurement are the lumen for luminous flux, and the candela for luminous intensity. Color is specified in terms of CIE recommended systems.[1]

4.0 Ambient and Physical Conditions

4.1 General

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment. Because lamps operate at high bulb temperatures, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before measurement. Most lamps can be easily cleaned with an alcohol-moistened paper towel.

4.2 Temperature

For practical purposes, an ambient temperature of 25 °C ± 10 °C (77 °F ± 18 °F) is recommended. Although temperature is not critical for incandescent filament lamps, care should be given to the requirements of the measurement instrumentation and the temperature coefficient of the detector.

4.3 Airflow

Airflow will not affect the life of incandescent lamps but will aid in cooling the life test area and the life test rack electrical components.

4.4 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Conditions

5.1 Wave Shape

If AC electrical power is used, the AC power source, while operating the test lamp, shall have a sinusoidal voltage wave shape such that the root-mean-square (RMS) summation of the harmonic components does not exceed 3% of the fundamental.

5.2 Voltage or Current Regulation

The DC voltage or current shall be regulated to within ±0.02%. The RMS voltage or current of the AC power source shall be regulated to within ±0.1%.

Measuring lamps on AC power will increase the measurement uncertainty compared to that for DC power.

The measurement circuits employed for incandescent filament lamps are shown in **Figure 5-1**. Usually a variable power source capable of providing DC voltages or current or AC voltages as required by the lamp is used. In either case, the lamp input voltage or current shall meet the requirements specified in this section.

5.3 Circuits and Lamp Connections

The method of connecting instruments into a DC circuit are shown in **Figure 5-1(a)**. The voltmeter, V, shall be connected at the base of the lamp socket. The switch is provided to remove the instrument from the circuit if it is necessary to determine a correction factor to compensate for its presence in the circuit. The switch should have low resistance and be rated at several times the actual current in the test.

Figure 5-1(b) shows the method of connecting an instrument or instruments into an AC circuit with capabilities of measuring wattage, current, and voltage. The voltmeter, V, and the potential element of the wattmeter, W, shall be connected at the base of the lamp socket. Corrections to compensate for the presence of the measuring elements in the circuit can be calculated from the impedance specification data provided by the manufacturer of the instruments, or they can be determined by using switches to remove instruments from the circuit.

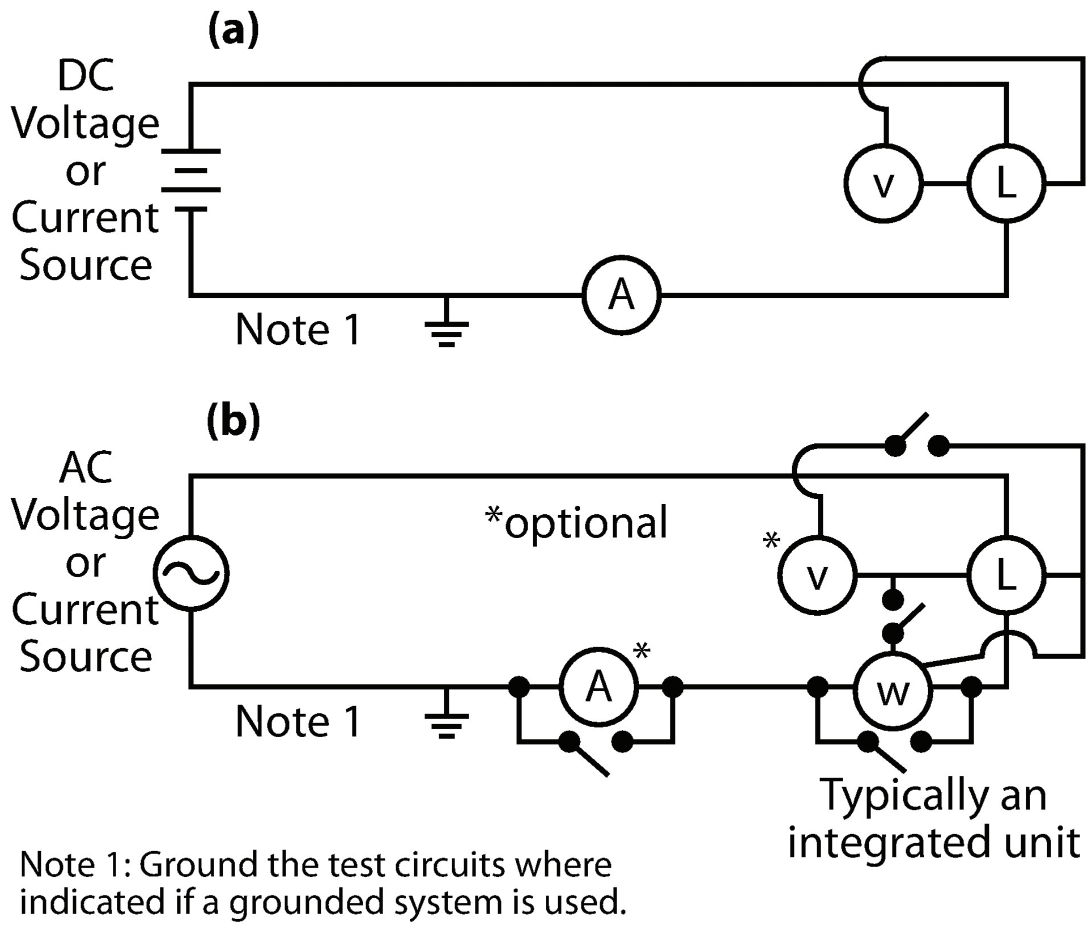

Figure 5-1. The methods of connecting instrument(s) into a DC circuit (a) and into an AC circuit (b).
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6.0 Lamp Test Procedures

6.1 Lamp Position

Lamp seasoning, pre-burning, and photometric measurements shall be done with the lamp in the same position. The operating position shall be as specified by manufacturer. For special application or test purposes, a position other than that specified by the manufacturer or determined by customary usage may be used and shall be noted in the test report. If no position is specified, the base up position shall be used.

6.2 Lamp Stabilization

6.2.1 Seasoning. Prior to taking initial measurements, all new incandescent lamps selected for test shall be seasoned per *ANSI/IES LM-54-20, Approved Method: Guide to Lamp Seasoning* (see to **Section 2**).

6.2.2 Pre-burning. Before any measurements are taken, the seasoned lamps shall be operated long enough to reach stabilization and temperature equilibrium. A period of one minute of continuous operation, or up to three minutes for tungsten-halogen types, is usually sufficient. However, it is always better to judge stability from periodic checks of light output, lamp voltage or current, or both, rather than elapsed time. When light output and/or lamp voltage (or current) become stable, the lamp is stable.

6.2.3 Determination of Stabilization Time. A generally accepted method for determining if an incandescent filament lamp is stable includes these steps:

Step 1. Take five measurements of the lamp light output at fifteen-second intervals (total time = 1 minute). This period is in addition to the recommended pre-burning time.

Step 2. Determine the maximum measured value, the minimum measured value, and the average value of the five consecutive measurements. Calculate the percent stability by dividing the difference between the maximum and minimum values by the average value of the five consecutive measurements.

Step 3. If the value calculated in *Step 2* does not exceed 0.25%, the lamp is considered stable.

Step 4. If the value calculated in *Step 2* exceeds 0.25%, the lamps are considered unstable and measurements in 15-second intervals have to be continued until the last five measurements meet the criteria in *Step 3*.

This method is strongly recommended. Variation in the acceptance criterion or in the number and duration of the measurement interval provides different levels of lamp stability, either more or less severe. Should a different method or set of criteria be used, it shall be so noted in the test report.

6.3 Electrical Settings

The standard method of taking photometric measurements of incandescent filament lamps shall be with the lamp stabilized and operating at the rated lamp voltage or lamp current. If the lamp is rated to operate over a range of voltage or current, the center value of that range shall be used. Any departure from this standard method shall be indicated in the test report.

6.4 Electrical Instrumentation

6.4.1 Instrument Tolerance. Instrument manuals provide values for the tolerance (accuracy) of the instrument, normally as a percentage of reading.[2,3]

The instrument tolerance for the measurement of electrical characteristics shall be as described in **Sections 6.4.1.1** and **6.4.1.2**.

6.4.1.1 DC Operation. The instrumental tolerance shall be: voltage ±0.05% or less, and current ±0.05% or less. The light output of incandescent lamps operated on DC power can vary when switching the electrical polarity to the lamp. This makes it necessary to label the negative and positive contacts of all lamps that can be inserted into sockets not keyed for polarity. All testing shall be done with the same polarity connections to the lamp.

6.4.1.2 AC Operation. The instrumental tolerance shall be: voltage ±0.5% or less, current ±0.5% or less, and wattage ±0.75% or less.

The reader is referred to NIST 1297 and JCGM 106:2012 for conversion of instrument tolerance limits to the measurement uncertainty.[2,4] The instrument(s) chosen shall have a specified accuracy adequate to ensure that these requirements are met.[2,3,4]

6.4.2 Impedance Limitations. Voltage input of the multifunction meter shall have input impedances greater than 1 megaohm (MΩ), and the current inputs shall have impedances less than 20 mΩ.

A test instrument connected in parallel with the lamp shall not draw more than 1 percent of the rated lamp current.

A test instrument connected in series with the lamp shall have an impedance such that the voltage across the instrument does not exceed 2% of the rated lamp voltage.

7.0 Photometric Test Procedures

7.1 Total Luminous Flux Measurements with an Integrating Sphere

7.1.1 Photodetector. For measurements using a photodetector that has a response function matching V(λ), the reader is referred to ANSI/IES LM-78-20 (see **Section 2**).

7.1.2 Spectroradiometer. The integrating sphere method gives total luminous flux of the test lamp with one measurement. Appropriate corrections shall be made unless substitution standards agree in spectral power distribution, physical size, and shape with the lamps under test. When the test lamps and the calibrating lamp are not of the same physical size, finish, degree of blackening, and shape, compensation for differences in self-absorption shall be made (see ANSI/IES LM-78-20; refer to **Section 2**).

Spectral lumen measurements[5] have many advantages over photodetector lumen measurements. There is no need for spectral mismatch correction due to non-flat sphere paint or the deviation of the photodetector response from V(λ), since every measured wavelength is individually calibrated.[6]

7.2 Measuring Luminous Intensity

Directional light output may be measured with one or more detectors at an appropriate distance from the lamp. Each detector shall have a relative spectral responsivity that approximates the V(λ) function (f_1' less than 3%). The recommended minimum distance is 10 times the lamp length in order to keep the error (caused by the assumption that the source is a point) less than 1%. Total luminous flux may be computed, if the flux-to-intensity ratio for the type of lamp is well established. (For detector sources of error such as spectral mismatch refer to ANSI/IES LM-78-20; see **Section 2**.) Other potential sources of error are:

- Stray light
- Errors associated with distance measurement
- Deviations from the inverse-square law (with respect to angular size of source)
- Errors associated with angular alignment of lamp and detector
- Materials that could reflect a significant amount of light, including the supports for the lamp or detector

7.3 Luminous Intensity Distribution

The distribution of luminous intensity around a lamp may be determined in a photometer similar to that used for normal luminous intensity measurements discussed in **Section 7.2** but provided with a means for varying the angles between the detector and the lamp axis.

Potential sources of error in addition to those listed in **Section 7.2** that require consideration are:

- Error in the scan angles
- Spectral errors due to spectral reflectivity of turning mirror(s)

The detector may be fixed and the lamp oriented at various angles in planes passing through the axis about which the lamp is moved. Readings are generally taken in 22.5-degree increments for symmetric spatial distributions. For asymmetric light output, the horizontal angular increment is reduced as needed to avoid inaccuracy resulting from sudden changes in distribution[7] for zonal luminous flux calculation.[6]

The photodetector system shall have a relative spectral responsivity that approximates the V(λ) function[8] (f_1' less than 3% unless spectral mismatch correction is applied).[5] There are numerous sources of error in making photometric measurements.[9] An assessment shall be made of these errors and their impact on measurement results. Significant sources of error should be either corrected or reported.

When lumen correction factors are used to correct measurement errors, they shall be well documented and analyzed.

7.4 Color Measurements

The measurement of color is specified in terms of the chromaticity coordinates, and color rendering indices as defined by CIE and practical guidance for implementation in IES documents.[10]

8.0 Test Report

The test report shall list all significant data for each lamp tested, together with performance data. The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamps. Typical items reported are:

a. Date and testing agency
b. Manufacturer's name and lamp designation
c. Type of test
d. Number of lamps tested
e. Rated electrical values for the lamp tested
f. Number of hours operated and operating cycle
g. Ambient temperature and temperature correction factor, if any
h. Lamp orientation during test
i. Seasoning and stabilization times
j. Type of standard used and correction factors, if any
k. Photometric method
l. Sphere diameter or test distance

m. Circuit conditions and fixed parameters
n. Measured light output (lumens and/or candelas) and electrical values (volts, amperes, watts) of each lamp, and group averages
o. Chromaticity data
p. Spectral power distribution
q. Flux/intensity ratio if lumens were calculated from intensity measurement data
r. Equipment used
s. Statement of uncertainty, if appropriate
t. Any deviations from standard operating procedures
u. Observations of any unusual behavior of lamps: e.g., range of measured lumen values when the stability reference limit is exceeded

REFERENCES

1 International Commission on Illumination (CIE). CIE 015:2018, Colorimetry, 4th ed. Vienna: CIE; 2018.

2 Taylor BN, Kuyatt CE. NIST Technical Note 1297, Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results. Washington, DC: National Institute of Standards and Technology; 1994.

3 Joint Committee for Guides in Metrology. JCGM 100:2008 (GUM 1995 with Minor Corrections), Evaluation of Measurement Data – Guide to the Expression of Uncertainty in Measurement. Paris: International Bureau of Weights and Measures (BIPM); 2008.

4 Joint Committee for Guides in Metrology. JCGM 106:2012, Evaluation of Measurement Data – The Role of Measurement Uncertainty in Conformity Assessment. Paris: International Bureau of Weights and Measures (BIPM); 2012.

5 International Commission on Illumination (CIE). CIE 53:1982, Methods of Characterizing the Performance of Radiometers and Photometers. Vienna: CIE; 1982.

6 Illuminating Engineering Society. ANSI/IES LM-20-20, Approved Method: Photometry of Reflector Type Lamps. New York: IES; 2020.

7 Illuminating Engineering Society. ANSI/IES LM-75-19, Guide to Goniophotometer Types and Photometric Coordinate Systems. New York: IES; 2019.

8 Illuminating Engineering Society. ANSI/IES LS-7-20, Lighting Science: Vision – Eye and Brain. New York: IES; 2020.

9 International Organization for Standardization (ISO) and International Commission on Illumination (CIE). ISO/CIE 19476:2014 (CIE S 023/E2013), Characterization of the Performance of Illuminance Meters and Luminance Meters. Vienna: CIE; 2014.

10 Illuminating Engineering Society. IES LM-15-93, IESNA Practical Guide to Colorimetry of Light Sources. New York: IES; 1993.

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